Race for Rare Earth between USA and China: Economic Implications

Submitted to:

[redacted]

Assistant Professor, Department of Economics

East West University

Submitted By:

Irfan Ahmed

Bachelors in Social Science (Economics)

Date of Submission: 31th January, 2025

Table of Contents

Table of Figures

Abstract

Global Competition for Global Earth elements (REEs) has become an important aspect of international economic and geopolitical dynamics as primary contenders with China and the United States. Reece, a group of 17 elements required for high -tech industries, clean energy and defense applications, are important for modern economies. China has installed a near-one in the REE market through strategic policies, state-supported investments and its supply chain, which is capable of controlling extraction, refinement and distribution. This dominance has given rise to the global supply chain weaknesses, as displayed as China's REE export ban in 2010, which has highlighted its ability to take advantage of REEs for geopolitical benefits.

In reaction, America has taken vital steps to reduce its dependence on the Chinese raze. These steps include reopening home mines like Mountain Pass, making an investment in technical improvements and creating strategic global partnerships to broaden an opportunity supply chain. However, demanding situations which include excessive manufacturing costs, environmental guidelines and constant dominance of China preclude American capability to reap self -reliance.

This paper examines the economic implications of the U.S.- China rare earth competition, analyzing China’s strategic approach, its impact on global trade, and the countermeasures adopted by the United States. In addition, it examines alternative sources of REE, including recycling, deep-C mining and emerging extraction techniques. This study shows the importance of international cooperation between the US, European Union, Japan and Australia, especially to create a variety of and flexible supply chains.

As the Global demand of the REE is increasing, especially in the renewable energy zones and defense industries, the geopolitical conflict over these important materials is expected to intensify. Understanding the economic influence of this race is necessary to design policies that promote supply chain stability, technological advancement and sustainable resource management.

Background

The global race to secure the rare earth elements (REEs) has intensified due to their crucial role in the high-tech industries, clean energy solutions and defense applications. The rare earths are a group of 17 elements, characterized by their unique properties like magnetism, luminescence and strength. They are essential for modern technologies including smartphones, wind turbines, electric vehicles (EVs) and advanced weaponry used in the military. In this global race, central competitors are the United States and China.

Through strategic investment and policy prioritization China has built a near-monopoly in REE production. Beijing has managed to develop a fully integrated supply chain by capitalizing on its resource dominance and it spans from extraction and processing to the production of advanced technologies. This dominance has allowed China to rule global REE trade through restrictive export policies and disrupted international markets that have triggered geopolitical tensions. For example, China's 2010 suspension of REE exports to Japan highlighted the vulnerabilities of other nations dependent on its supply.

The United States has undertaken efforts to revitalize its domestic REE industry. These initiatives include legislative measures, investments in alternative supply chains and international collaborations. Nevertheless, barriers like high production costs, environmental challenges and China's established market dominance remain significant.

This paper examines the economic implications of the U.S. - China competition over rare earths. It explores the rise of China as a dominant REE producer, strategic motivations behind different policies, impact on global markets and supply chains. Furthermore, it analyzes U.S. responses and their attempt to mitigate dependency on Chinese resources. By analyzing these dynamics, the study contributes to the broader understanding of critical mineral's role in shaping economic power and geopolitical relations in the 21st century.

Chapter 1: Introduction to Rare Earth Elements (REEs)

Rare Earth Elements (REEs) consist of a group of 17 chemicals elements which include yttrium, scandium, and the 15 lanthanides, all of which are necessary for a myriad of advanced technologies. All of these materials are relatively plentiful on Earth’s crust, despite being designated as “rare”. Nevertheless, all of these materials are sporadic and sparsely concentrated, which makes the extraction process economically arduous. REEs possess qualities such as, magnetism, luminescence and electrochemical properties – which make them crucial in numerous industries.

Neodymium and dysprosium, for instance, are invaluable in the manufacture of powerful permanent magnets utilized in wind turbines and electric vehicles motors. Likewise, in order to produce phosphors in energy-efficient lighting and display technologies, materials such as europium and terbium are essential. In addition, REEs are imperative in the production of catalysts in petroleum refining, leading to cleaner fuel production.

Challenges in Mining and Refining REEs

The extraction and processing of REEs prove to be significantly environmentally and economically challenging. Mining operations involve excavation which contributes to soil erosion and natural habitat destruction. Since the refining process is chemically intensive, accumulating considerable amounts of radioactive byproducts such as thorium and uranium, it adversely affects the environment and poses health risks. The mining process generates around 13 kg of dust, 9,600-12,000 cubic meters of waste gas, radioactive remnants, and 75 cubic meters of toxic water, for every one ton of rare earth extracted.

The environmental concerns have led to austere regulations in numerous countries, which have raised complexities and costs of REE production. As a consequence, utilizing its low labor costs and lenient environmental regulations China has emerged as a dominant, near-monopoly in the mining and refining sectors of global REE market. Recent estimates have shown that 65% of global REE production and approximately 85% oof refining capacity is controlled by China.

Chapter 2: China's Strategic Dominance in the REE Market

Historical Development of REE Industry

China's dominance in the REE market is owing to several decades of deli berate and strategic planning. In the 1980s, Chinese government initiated extensive investment in their extraction and processing, as they realized the strategic importance of REEs. Through integrating numerous small mining operations into state owned enterprises, China was able to achieve economies of scale and improved control over their operations. Government policies, such as subsidies and export restrictions, had favored the integrated mines and limited foreign competition and amplified domestic industry growth.

By the 2000s, China had climbed to be the global leader in both the REE mining and refining industry. Chin's infrastructural and technological investments throughout the decades had enabled them to dominates the supply chain, allowing them to be the primary supplier of REEs in the global market. This dominance had further been cemented through China's concentration in the downstream industries, technological leadership and higher security in the supply chain, facilitating them to secure greater value from REE resources.

Supply Chain Control and Strategic Policies

China's control over the international REE supply chain had resulted from thorough and extensive policies, including; export controls, environmental regulations and production quotas. These policies had been strategically designed to ensure domestic availability of resources while restricting supply of REEs to foreign competitors. For example, export quotas have been imposed by China to limit the amount of exported REEs, allowing them increased control in international prices and global supply.

Furthermore, relative to other countries China has been able to increase efficiency and cost control in processing and refining REEs, through heavily investing on technological advancements. These investments have given them a competitive edge, enabling them to supply 85% pf global REEs.

Geopolitical and Economic Leverage

China's unyielding control in the global REE market has provided them with considerable geopolitical and economic leverage. In 2010, China restricted REE supply to Japan during a territorial dispute, this skyrocketed the prices and accentuated the significance of REEs as geopolitical tools. This had reinforced Chin's ability to influence the REE market to further their economic and political gains, alluding the strategic importance of REEs in intergovernmental relations.

Additionally, the dominance and concentration of REE production in China has raised concerned for global industries relying these materials. The concerns include external shocks such as supply chain vulnerabilities, leading to other nations alternative sources and develop measures to decrease reliance on Chinese REEs.

Chapter 3: U.S. Efforts to Reduce Dependency on China

The U.S. Strategy for Diversifying the REE Supply Chain

The United States has implemented several strategies to diversify the supply chain of REE and reduce the dependency on Chinese sources, in response to their dominance in the global supply chain. These measures include reinstating domestic mining operations, such as Mountain Pass Mine in California and increasing investments on extraction technologies. In 2018, the Mountain Pass Mine had reopened which was previously closed due to environmental concerns. This was done with the objective of decreasing reliance on foreign REE sources.

Moreover, the U.S. government had improved REE extraction and processing technologies by increasing investment on research and development, devoting to more economically and environmentally sustainable means of production. These policies were aimed to expand production capabilities and reduce foreign commodity dependence.

Technological Innovations for Efficient REE Production

Technological innovations are being developed so that REE extraction is carried out more efficiently and in an environmentally sustainable manner. Innovations, namely, bioleaching employs microorganisms to extract REEs.

Chapter 4: Recycling and Alternative Sources of Rare Earth Elements (REEs)

The Promise of REE Recycling

Recycling REEs poses a promising approach to alleviate environmental and economic challenges entailed with primary mining. As the demand for REEs soars, the necessity for sustainable recycling practices becomes ever-increasing, especially in renewable energy sources.

These challenges are addressed through innovating recycling technologies. For example, the QUILL research center at the Queen's University Belfast has formed measures which employ ionic liquids to efficiently recover REEs from discarded magnets used in devices such as electric vehicles and wind turbines. These measures have minimal environmental impacts relative to more traditional techniques.

In spite of these innovations, myriads of obstacles persist. The expenses of recycling often outweigh the cost of primary mining, making mining economically unviable in many cases. In addition, the complexities and noxious nature of REE extraction from discarded products raises further environmental and health concerns. Such as, recycling is imbued with hazardous chemicals and high temperatures, which may potentially lead to contamination if not managed with caution.

Contemporary research is fixating on developing more efficient and environmentally sustainable methods of recycling with the objective of prevailing these challenges. Breakthroughs in green chemistry and renewable energy technologies are advocating for a circulating economy approach, aiming to reduce dependency on primary sources and mitigate environmental impact.

Alternative Sources and Technologies

Along with recycling, alternative sources of REEs are being developed to variegate supply chains and decrease dependency on primary mining. One lucrative pathway is the extraction of REEs from waste streams, including coal ash and industrial residue such as red mud. These sources would potentially supply REEs without environmental impairments associated with traditional mining.

In addition, another innovative method involves phytomining, where plants called hyperaccumulators absorb high quantities of minerals from the soil. Thereafter, the plants can be harvested and process to extract REEs from them. Even though still in developmental stages, phytomining provides a relatively less expensive and environmentally pernicious substitute to traditional mining practices.

Deep-sea mining shows a promising avenue to potentially source REEs. The ocean floor contains abundant deposits of polymetallic nodules permeated in REEs. Nonetheless, the environmental implications of deep-sea mining is a concerning factor, and large-scale implementation may not be carried out in the foreseeable future and may be years away.

Countries such as Canada, Australia and European Union members are investing in environmentally friendly mining and REE processing practices. These practices include establishing alternative supply chains and abating dependency on dominant REE producers, such as China. For instance, the U.S.-led Minerals Security Partnership had promoted a new rare earths project in Brazil, aiming to diversify invaluable mineral supply from China's control.

In summation, whilst recycling and alternative avenues of REEs offer prospective solutions to supply chain constraints, all of them contain their own set of drawbacks. In progress research and international collaboration are crucial to developing economically sustainable and viable means of REE recovery and mining.

Chapter 5: Geopolitical and Economic Implications of the REE Race

Strategic Importance of REEs

Rare earth elements (REEs) are salient to various high-tech applications, which include electronics, renewable energy technologies and defense systems. Their unequaled characteristics make them essential to produce components such as, magnets, batteries and other valuable constituents. As a consequence, control over REEs has become a matter of geopolitical and economic concerns.

China has gained economic leverage due to their dominance in the REE market. China has positioned itself to be a predominant player in any global trade negotiations and geopolitical conflicts, through controlling 70% of world's REE production and around 90% of processing capacity. This has raised concerns in many nations regarding their stability and security of supply chains.

A disruption or impediment in the REE supply chain of United States and its allies could result in drastic consequences in the industries such as defense, electronics and renewable energy. For instance, insufficiency in REEs could halt the production of electronic vehicles, wind turbines and advanced military equipment – leading to a potential threat to national security and economic stability.

National Security and Economic Imperatives

Reducing the dependency on Chinese REEs is not only an economic necessity for United States but also a subject of national security. An unyielding and diversified supply chain is crucial for maintaining military and technological edge. This has encouraged considerable investments in domestic production, international collaboration and recycling to capture alternative REE supplies.

Projects such as reopening the Mountain Pass Mine in California and development of new processing facilities aim to boost domestic REE production. Additionally, the U.S, has partnered up with Japan and European Union to invest in new technologies and diversify supply chains, potentially leading to fall in China's monopoly in the REE market and ensuring a more equitable allocation of critical materials.

Besides, in order to further diversify the supply chains the U.S. has formed strategic partnerships with countries bestowed with abundant REE reserves. These measures would alleviate the risks of an over-dependency on a single supplier, as well as promoting sustainable mining and processing practices.

Geopolitical Realignments

The international REE competition has led to the formation of alliances and geopolitical realignments. For instance, the Quadrilateral security Dialogue (Quad) comprised of the United States, Japan, Australia and India, has ventured into securing alternative sources of REEs. These collaborations are changing the patterns of global supply chains, with the potential to ameliorate its fortitude and sustainability.

Chapter 6: Future Trends and Challenges in the REE Market

Diversification and Sustainability Efforts

The importance of diversifying global supply chains to ensure sustainability has been ever increasing for nations around the world, as the global demand for REEs continually elevates. For years, China has been the dominant producer and processor, leaving the world highly dependent on their supply. However, with the rise in geopolitical tensions between United States and China, countries are seeking for alternative sources of REE supplies.

One of the most significant campaigns was European Union's (EU) Action Plan on Critical Raw Materials. This initiative was carried out with the aim of reducing EU's dependency on China's supply of imports and increasing secured access to critical rare materials such as REEs. This also included a plan to recycle more REEs, enhance the mining technologies and achieving more efficient processing techniques. EU has shown interests in extraction within its own borders; in particular, interests have been raised on mining projects in Greenland which is home to large REE deposits. However, political and environmental challenges persist which make such ventures formidable.

Technological Advancement and Innovations

Technological advancement and the future of REE is interconnected as it can amplify the efficiency and environmental sustainability of extractions and processing. The requirement for such innovations is crucial as the demand for REE will likely increase with the world gradually shifting towards renewable energy and advanced electronics.

One of the most promising innovations in the area is bioleaching, a process that is used to extract metals from ore through microorganisms. This technique requires fewer toxic chemicals compared to traditional chemical leaching which minimizes environmental pollution and makes it a more environmentally friendly method. Bioleaching has been used to successfully extract metals like copper and gold, making researchers eager to use it for REE extractions. This could offer a more cost-effective way to recover rare earth elements from the earth's crust, especially the ones that are low-grade-deposit and are economically not feasible for conventional mining.

Another area of improvement involves the formation of alternative materials to reduce dependency on REEs, particularly in technologies like magnets and batteries. For instance, researchers are experimenting with different ways to create permanent magnets that do not require neodymium or dysprosium, two most widely used REEs. Using alternatives like iron, cobalt or nickel may allow them to generate high performance magnets that are abundant as well as less expensive compared to REE-based counterparts. Similarly, in terms of batteries, researchers are examining alternatives to the lithium-ion batteries that are dependent on REEs like sodium-ion batteries that can provide more sustainable solutions.

Improvements to recycling methods for REEs can be considered as another crucial technological development. The traditional recycling process for REEs is costly and energy-intensive, but the improvements in methods like ionic liquid-based and solvent extraction are making it easier to extract these valuable materials from end-of-life products. Through increasing the efficiency of these processes, the environmental effect of REE mining can be decreased to some degree and be able to generate a sustainable, circular supply chain.

Global Collaboration to Counter China's Influence

Since the geopolitical significance of REEs is growing, the need for international collaboration is becoming more crucial to ensure a stable and diversified supply of these materials. The United States, European Union, Japan and other countries have realized the strategic importance of reducing reliance on China, which is controls a large portion of world REE supply and processing. In response these countries are collaborating to share knowledge, methods, resources, technologies to counter Chinas dominance in the REE market.

For instance, to diversify REE supply chains and to develop technologies, the U.S. Department of Energy has partnered with Japan and the EU. This collaboration aims to improve mining and processing technologies, enhance the recycling of REEs and create alternative sources of supply. Along with this, the U.S has entered into bilateral agreements with different countries like Australia, Brazil and Canada to stabilize their access to REE reserves and decrease the strategic dominance of China in international supply chains.

Another example of a multilateral partnership is The Quadrilateral Security Dialogue (Quad), comprising the United States, Japan, India, and Australia, focusing on securing alternative REE sources. It has highlighted the requirement of diversification of the supply chains for critical raw materials, including REEs, and has inspected possibilities for joint investments in mining and processing projects. These collaborations intent to reduce the global dependence on China and promote regional stability as well as security.

Furthermore, the creation of a global REE market outside of China has the possibility to create new trade agreements and alliance. Countries that own significant REE reserves, like Brazil, Vietnam and Russia are prominently seen as the key players in the world REE supply chain. By collaborating with these nations, the U.S and its allies can ensure availability to significant resources while reducing China's impact over the market.

Breakdown of Global Rare Earth Exports (2008-2018)

Country	Export Volume (metric tons)	Share (%)	Export Value (millions of US$)	Share (%)
China	407,886.6	42.3	8,112.2	46.3
USA	89,467.1	9.3	953.6	5.4
Malaysia	87,696.1	9.1	942.4	5.4
Austria	87,055.1	9.0	867.8	5.0
Japan	68,412.9	7.1	2,172.6	12.4
Rest of World	223,172.7	23.2	4,467.7	25.5

Source: UN Comtrade Database

FIGURE 01

Starting from 2008-2018, majority of rare earth export was dominated by China, contributing 42.3% of global volume (407,886.6 metric tons) and 46.3% of export value ($8.1 billion), which shows its stronghold in both production and processing. Other countries, namely: U.S., Malaysia, and Austria followed with similar export volume of 9.3%, 9.1% and 9%, respectively – significantly lower that China.

Despite exporting only 7.1% of volume, Japan accounted for 12.4% of total value (**$2.17 billion**). The rest of the world contributed 23.2% of exports (223,172.7 metric tons) and 25.5% of value ($4.47 billion). Thus, the data verifies China's dominance in the rare earth trade, while other countries much play smaller roles.

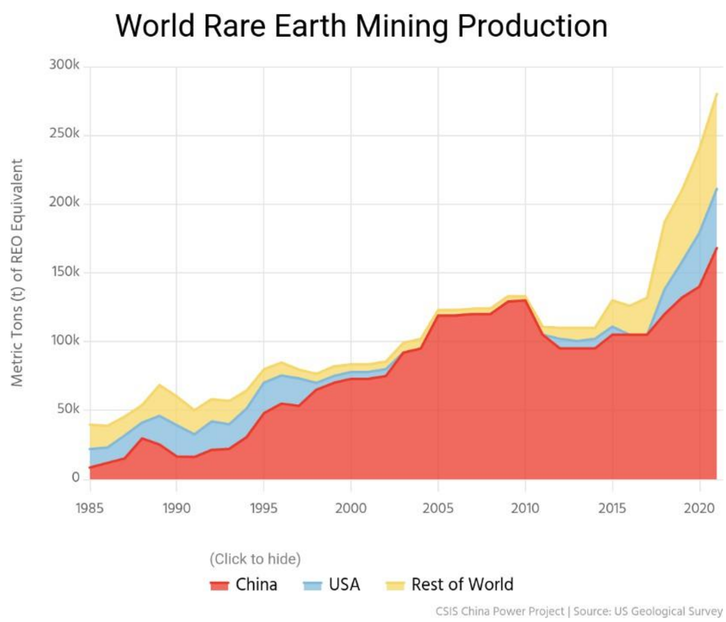

FIGURE 02

The graph shows global rare earth mining production from 1995 to 2020, highlighting the contribution of China, the USA and the rest of the world. According to the graph, it can be seen that China’s share (represented in red) has increased significantly since the 1980s, rapidly taking the dominant position by the early 2000s. And despite some relatively stable fluctuations, it managed to maintain its leadership after the 2010.

On the other hand, the USA (in blue) was a primary controller until the early 2000s, and has seen a reduction of output since the last 10 years. However, production has rebounded since 2015, which could highlight renewed efforts to reduce Chinese supply chain dependency.

Comparatively, the rest of the world (in yellow) played a rather trivial but growing part, with production remaining relatively stable. This indicates diversification in rare earth supply, likely driven by rising global demand and geopolitical concerns.

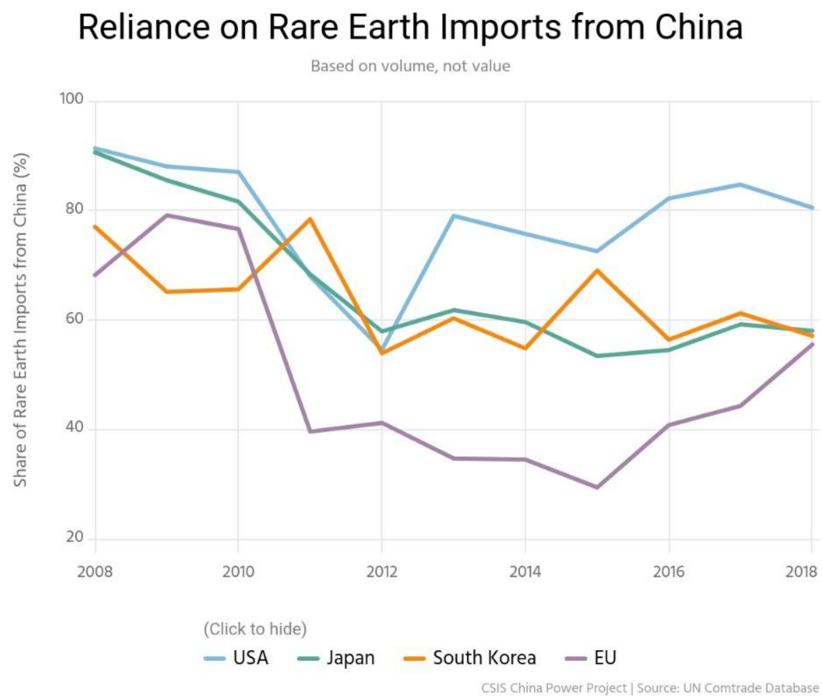

FIGURE 03

This graph shows the share of rare earth imports from China for the United States, Japan, South Korea and the European Union from 2008 to 2018, measured by volume rather than value. Initially, all four regions trusted China a lot, in which import shares were more than 80%. However, over time, the dove who trends appeared.

The USA (Blue Line) maintained a high dependence, experienced a brief decline around 2012, followed by an increase. Japan (Green Line) gradually reduced its dependence, especially after 2010, due to efforts to diversify the supply. South Korea (Orange Line) showed up and down with a sharp decline around 2012, followed by a recovery. The most significant decline in the European Union (Purple Line) fell by 40% by 2012, but later a gradual increase was observed.

US Defense Applications of Rare Earths

Element	Communication	Electric Motors	Guidance and Control	Targeting and Weapons
Dysprosium		•	•	
Europium	•			•
Lanthanum	•			
Lutetium	•			
Neodymium	•	•	•	
Praseodymium		•	•	
Samarium		•	•	
Terbium		•	•	•
Yttrium	•			•

Note: There are also numerous rare earths used for electronic warfare technologies.

Source: Congressional Research Service

FIGURE 04

Table describes the defense programs of rare land elements in four major categories: communication, electric motors, guidance and control and direction and weapon. These elements are essential for advanced military technologies, contributing to the influence of various defense systems, durability and efficiency.

Among the listed elements, neodium, presidomium and Samaryium play an important role in electric motors, which are important for propulsion systems and high performance actuators in defense applications. Discipline and Terbio also contribute to electric motors, leading to an increase in their performance under extreme conditions.

Rare land elements such as Europeans, Lanthanum, Lutetium, Neodemios and Yatrium are used in communication techniques, essential for safe military operation, data transmission and satellite systems. Discipline, neodymium, passsodemium, summerium and terbio are also involved in orientation and control systems, which are crucial for precision direction, missile orientation and drone navigation.

Finally, dysprosium, neodymium, pasodymium, samarium, terbio and yttrio are used in segmentation and weapon systems, which include advanced missile systems, laser direction and other high precision military technologies. This highlights the strategic importance of rare lands in maintaining defense capabilities, making their supply chain a crucial geopolitical concern.

Finally, dysprosium, neodymium, pasodymium, samarium, terbio and yttrio are used in segmentation and weapon systems, which include advanced missile systems, laser direction and other high precision military technologies. This highlights the strategic importance of rare lands in maintaining defense capabilities, making their supply chain a crucial geopolitical concern.

Reliance on Chinese Rare Earth Metals and Alloys (2018)

Economy	Imports from China (metric tons)	Total Imports	% of Imports from China
EU	869.2	882.2	98.5
USA	417.6	438.6	95.2
South Korea	86.1	94.6	90.9
Japan	4,233.4	8,729.2	48.5

Note: Excludes oxides and other compounds

Source: UN Comtrade Database

FIGURE 05

The table presents data on dependence on large economies on Chinese rare soil metals and alloys in 2018 by showing the amount of rarely soil imports from China, the total import and the percentage of imports taken from China. The economies listed include the EU (EU), USA (US), South Korea and Japan.

The European Union (EU) had the very best dependence on China, with 98.5% of imports from the uncommon land of China and totaled 869.2 heaps of a complete of 882.2 heaps. This highlights the almost entire dependence of China's EU for these essential materials, making the deliver chain very at risk of issues in Chinese exports.

The United States (USA) also showed a high dependence and imported 417.6 tons of China, representing 95.2% of total imports (438.6 lots). This sizable addiction emphasizes the strategic importance of ensuring alternative assets to lessen publicity to feasible geopolitical risks.

South Korea imported 86.1 heaps of uncommon soil metals from China, representing 90.9% of general imports (94.6 tons). Although the full volume is lower in comparison to the EU and the USA, South Korea still suggests a strong dependence on China for those materials, that's crucial to electronic and technological industries.

Unlike other economies, Japan showed a more diverse import pattern. Although it imported 4233.4 tons of China, this represented only 48.5% of total imports (8 729.2 tons). Japan's lowest percentage confidence has made an effort to diversify the rare supply of the soil, possibly obtaining alternative suppliers or investing in domestic production and recycling.

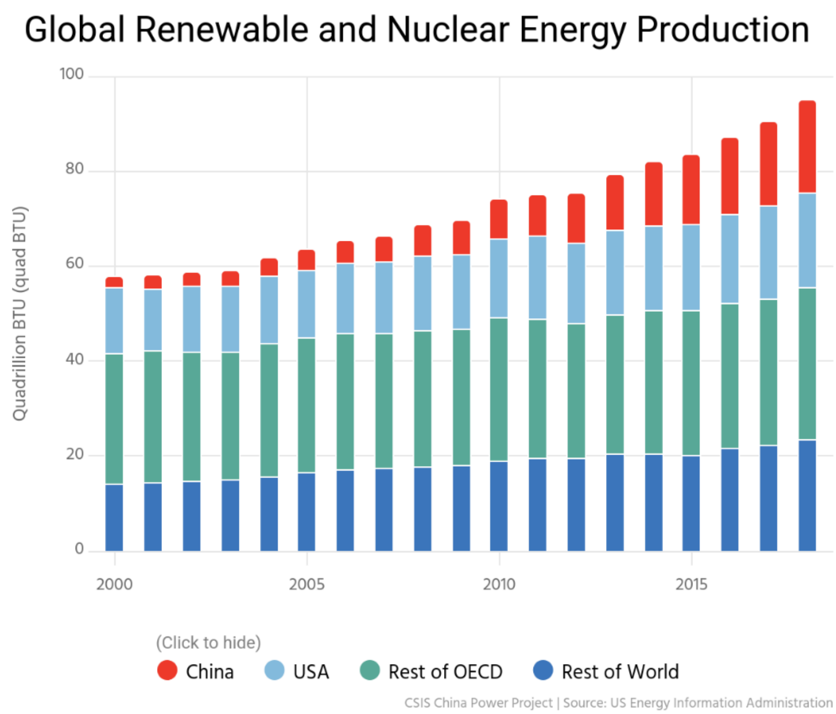

FIGURE 06

China's participation in global rare land production decreased from 97.7% in 2010 to 62.9% in 2019. This reduction is the result of increased US mining operations and elsewhere, as the mountain passage mine.

Companies such as MP Materials are working to increase processing capabilities internally rather than relying on China for the refining of rare land materials. Similarly, Joint ventures such as Lynas Corporation (Australia) and Blue Line Corporation intend to establish US processing facilities, reducing dependence on China.

The Japanese government, through its body, is supporting these initiatives, hoping to reduce Japan's dependence on Chinese imports of rare land. The goal is to reduce Japan's dependence to less than 50% to 2025.

The demand for rare lands, especially driven by the growing renewable energy sectors and electric vehicles, is pressing global supply chains. The growing consumption of these materials in China is pressing to become a liquid importer, weakening its domain in industry.

As China's domestic needs increase, its ability to control the global market can weaken, allowing emerging competitors to gain a position in the rare land industry. The change in global supply dynamics, along with the growing demand for these materials, can level the field of play, reducing China's market power.

Conclusion

Global competition on rare earth elements between the United States and China has significant economic, geopolitical and technological implications. China has leveraged decades of strategic investment, control of the policy-oriented supply chain and economic production to establish an almost monopoly on the REE market. This condition gave China a significant economic leverage and affected global supply chains and international connections.

In response, the United States has started different strategies to reduce the dependence on Chinese REEs, including revitalizing domestic mining operations, promoting technological innovations for recovery and recycling and promoting international cooperation. However, challenges such as high production costs, environmental concerns and China's entrenched market position continue to make these efforts difficult.

Future trends indicate a growing emphasis on diversification, sustainability and innovation in REEs production and processing. Recycling initiatives, alternative sources of mining and technological advances, such as REE bioleaching and substitutes, offer potential solutions for supply chain vulnerabilities. In addition, global alliances, including partnerships between the US, Japan, EU and Australia may assist to combat China's dominant position and create a more resilient, diverse REE market.

Although China's market control remains strong, change in global dynamics - as the largest domestic demand in China and increased competition - can gradually reduce the monopoly. A balanced and diversified REEs supply chain will be fundamental to ensuring economic stability, technological progress and geopolitical security in the coming years.

References

1. Paulick, H., & Machacek, E. (2017). *The global rare earth element exploration boom: An analysis of resources outside of China and discussion of development perspectives*. *Resources Policy, 52*, 134–153. https://doi.org/10.1016/j.resourpol.2017.02.002

2. Kalantzakos, S. (2020). *The race for critical minerals in an era of geopolitical realignments*. *The International Spectator, 55*(3), 1–16. https://doi.org/10.1080/03932729.2020.1786926

3. Mancheri, N. A. (2015). *World trade in rare earths, Chinese export restrictions, and implications*. *Resources Policy, 46*, 262–271. https://doi.org/10.1016/j.resourpol.2015.10.009

4. China Power Team. (n.d.). *How China's dominance in rare earths impacts global supply chains and geopolitics*. *China Power, Center for Strategic and International Studies (CSIS)*. Retrieved from https://chinapower.csis.org/china-rare-earths/#:~:text=support%20basic%20industries.-,Deep%20Reliance%20on%20China,pursuit%20of%20its%20political%20objectives

5. Texas Comptroller of Public Accounts. (2021). *Rare earth elements: Supply chain deep dive*. Retrieved from https://comptroller.texas.gov/economy/economic-data/supply-chain/2021/rare-earth.php

6. International Energy Agency (IEA). (2023). *Global EV outlook 2023*. Retrieved from https://www.iea.org/reports/global-ev-outlook-2023

7. Business Insider. (2024, November). *Rare earths and the tech war: How China's dominance impacts AI and chips*. Retrieved from https://www.businessinsider.com/rare-earths-tech-war-explainer-chips-ai-china-us-2024-11

8. BBC News. (2019, May 29). *Rare earth elements: Why China is cutting exports to the US*. Retrieved from https://www.bbc.com/news/world-asia-48366074

9. Seaman, J. (2019). *Rare earths and China: A review of changing criticality in critical materials*. *Institut Français des Relations Internationales (Ifri)*. Retrieved from https://www.ifri.org/sites/default/files/migrated_files/documents/atoms/files/seaman_rare_earths_china_2019.pdf

10. World Economic Forum. (2019, June). *Rare earths: The new battlefront in the US-China trade war*. Retrieved from https://www.weforum.org/stories/2019/06/rare-earths-are-the-new-battlefront-in-the-us-china-trade-war-but-what-precisely-are-they